

PROJECT DEVELOPMENT PHASE

India's Agricultural Crop Production Analysis (1997-2021)

A Tableau Dashboard built as part of the SmartBridge Data Analytics using Tableau Internship

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Department	Artificial Intelligence and Machine Learning
Internship	Data Analytics using Tableau
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1. Dashboard Visualizations

The finished dashboard contains nine visual elements. The table below explains the purpose of each and the key insight it delivers.

Visualization	Purpose	Insights
Total Production KPI	Show the national aggregate production figure at a glance.	Total production stands at 331.00B units across all crops and years, giving an immediate headline figure.
Total Yield KPI	Show the national aggregate yield figure at a glance.	Total yield of 27.43M reflects overall productivity per cultivated hectare across India.
India Map	Plot state-wise production geographically using color intensity.	Southern states, especially Kerala and Tamil Nadu, show the darkest shading, indicating the highest production.
Year-wise Production	Track how national production has moved year by year.	Production shows a broadly rising trend from 1998 to 2021, with a few years of flatter growth.
Agricultural Yield Trend	Track productivity per hectare over the same 24-year period.	Yield improved steadily through the study period with small year-to-year fluctuations.
State-wise Cultivated Area	Compare how much land each state farms.	Uttar Pradesh, Madhya Pradesh, and Rajasthan have the largest cultivated areas.
Production Distribution by Unit Type	Break total production down by measurement unit.	Tonnes make up the overwhelming majority of production, with bales and nuts as smaller shares.
Leading Agricultural States	Rank states by their total production.	Kerala leads by a wide margin, followed by Tamil Nadu and Karnataka.
Season-wise Production	Show how production splits across cropping seasons.	The Whole Year season contributes the overwhelming majority of production, with Winter and Summer contributing much smaller shares.

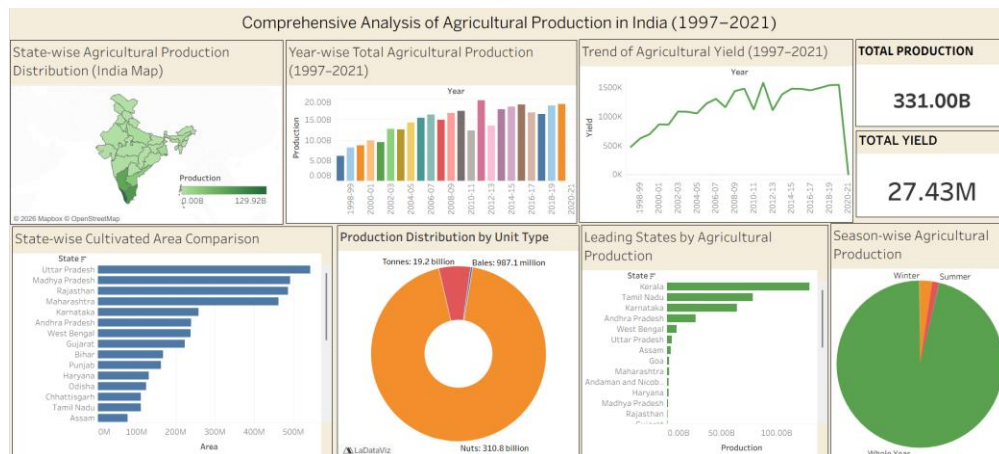


Figure 2: Full dashboard view showing all nine visualizations arranged together.

2. Dashboard Actions

To make the dashboard genuinely interactive rather than a static report, a dashboard action was configured on the India map so that clicking any state filters every other chart on the dashboard to that state's data.

Dashboard Actions and Cross Filtering

A 'Filter' action was set up with the India map as the source sheet and every other chart on the dashboard as target sheets. When a state is clicked, Tableau automatically applies that state as a filter across the whole dashboard, which is what is commonly called cross filtering.

Interactive Analysis and Keep Only

The screenshot below shows the state of Kerala selected on the map, with the right-click context menu displaying 'Keep Only' and 'Exclude' options alongside the tooltip that reports Kerala's exact production figure (129.92B). Selecting Keep Only locks the whole dashboard to Kerala's data only, while Exclude would remove Kerala and recalculate every chart for the remaining states.

Dynamic Filtering and What Happens When Kerala Is Selected

Once Kerala is selected, the Total Production and Total Yield KPI cards recalculate to show Kerala's numbers alone, the cultivated area bar chart collapses to a single Kerala bar, the donut chart re-scales to show Kerala's unit-type split, the leading states chart isolates Kerala, and the season-wise pie chart updates to reflect only Kerala's seasonal production. Every visualization updates automatically and simultaneously - the viewer does not need to manually re-filter each chart one by one.

Benefits

- Lets a viewer drill from a national overview down to a single state in one click.
- Removes the need for separate state-level dashboards, since one dashboard serves every state.
- Makes the dashboard feel like a real analytical tool rather than a fixed set of pictures.

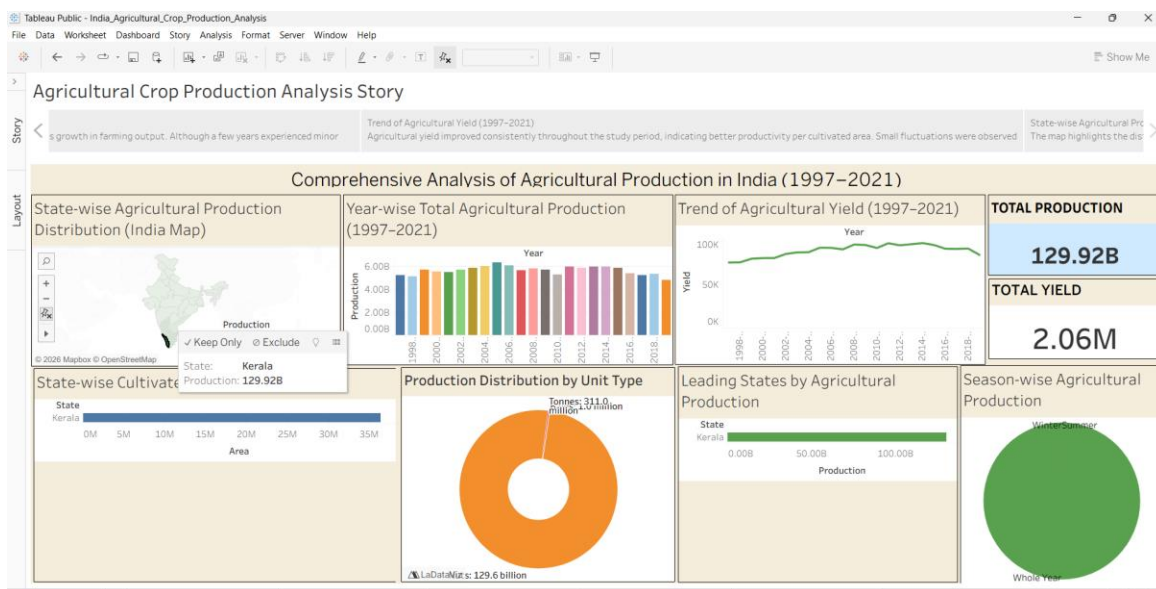


Figure 3: Dashboard action triggered by selecting Kerala on the India map, showing the Keep Only / Exclude options and Kerala's production tooltip.

3. India Map Development

The India map is the centerpiece of the dashboard and required the most care to build correctly, since Tableau does not always recognize Indian state names automatically.

Map Layers and State Boundaries

The map uses Tableau's built-in state-boundary layer for India, with each of the country's states and union territories outlined individually. State names in the dataset were matched against Tableau's geographic role for 'State/Province' so every polygon rendered without unknown or unplotted locations.

Color Encoding and Production Scale

Color is encoded on a single-hue green gradient tied to the Production measure, running from a light green (0.00B) to a dark green (129.92B), as shown in the legend. This lets a viewer instantly identify high-production states purely by how dark they appear, without needing to click into each one.

Tooltips and Interactivity

Hovering or clicking any state shows a tooltip with the exact state name and production value - for example the screenshot below shows Uttar Pradesh with a production figure of 4.58B. This gives an exact number to back up the visual color-based impression, and the same click also drives the dashboard action described on the previous page.

How Users Analyze State-wise Production

A user can scan the map for the darkest regions to spot leading states at a glance, hover over any state for its precise figure, or click a state to filter the entire dashboard down to that state's detailed breakdown - combining a quick visual overview with an exact-value drill-down in a single chart.

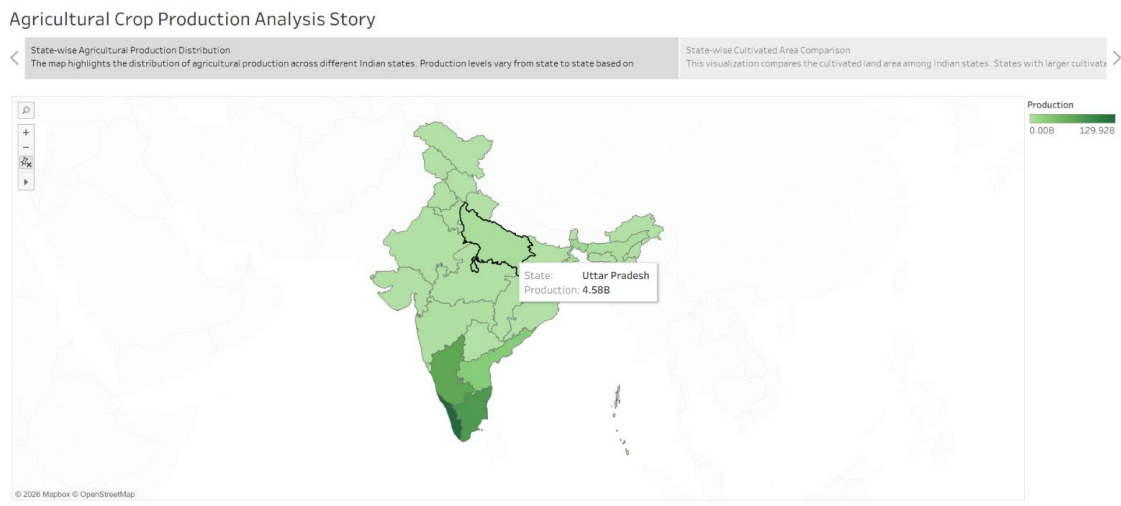


Figure 4: India map (Story view) showing Uttar Pradesh selected with a production tooltip of 4.58B.

4. Story Development

Purpose of the Tableau Story

Beyond the interactive dashboard, a Tableau Story titled 'Agricultural Crop Production Analysis Story' was built to guide a first-time viewer through the data in a logical sequence, rather than leaving them to explore nine charts with no starting point.

Navigation

The story uses Tableau's built-in navigator strip at the top, letting a viewer move forward and backward between story points using the arrow controls, or jump directly to a specific point by clicking its caption in the strip.

Story Points

Each story point captures one dashboard view along with a short caption summarizing the insight for that view - for example the state-wise production map is paired with a caption explaining that production levels vary state to state, and the cultivated area chart is paired with a caption comparing land area across states. This mirrors the two figures shown earlier in this document (Figures 3 and 4), which were both captured from within the Story interface.

Interactive Presentation

Because each story point is built on top of the live dashboard, the underlying charts remain interactive even while presenting - a viewer can still click a state within a story point and see the cross-filtering behavior described on Page 8, blending a guided narrative with hands-on exploration.

Each Story Point Represents One Business Insight

The story was structured so that every point answers one of the business questions defined on Page 5 - one point per insight, such as the state with highest production, the yield trend over time, or the leading contributing season - so that by the final story point, the viewer has been walked through the whole analysis end to end.

5. Publishing to Tableau Public

Publishing Steps

Once the dashboard and story were finalized, the workbook was published from Tableau Desktop using Server > Tableau Public > Save to Tableau Public, signing in with the team's Tableau Public account and naming the workbook 'India_Agricultural_Crop_Production_Analysis'.

Public URL

After publishing, Tableau Public generated a shareable public URL under the profile shaik.imamkhasim, in the format public.tableau.com/app/profile/shaik.imamkhasim/viz/India_Agricultural_Crop_Production_Analysis, which can be opened by anyone without needing a Tableau license.

Verification and Browser Testing

The published link was opened in a browser to confirm that the dashboard loaded correctly outside of Tableau Desktop - every KPI card, chart, and the India map rendered exactly as designed, and the dashboard action (clicking a state) was re-tested directly on the live public page to confirm it still worked after publishing.

Dashboard Accessibility

The published version is fully accessible online, requires no sign-in for viewers, and retains all interactivity - filters, dashboard actions, and story navigation - exactly as it behaves in Tableau Desktop.

The dashboard was successfully published to Tableau Public and verified as fully functional and accessible from a standard web browser, as shown below.



Figure 5: The dashboard live on Tableau Public, viewed in a browser at the published public URL.

6. Testing During Development

Dashboard Validation

The overall dashboard was checked at multiple stages of development to confirm every container resized correctly, no chart overlapped another, and all nine visualizations loaded without errors when the workbook was reopened.

KPI Validation

The Total Production and Total Yield KPI values were manually recalculated in Excel using SUM and AVERAGE formulas on the raw dataset and compared against the numbers shown on the KPI cards to confirm they matched exactly.

Map Validation

Every state in the dataset was checked against the India map to confirm it plotted in the correct location with no unrecognized or unplotted states, and tooltip values were spot-checked against the underlying data.

Chart Validation

Each of the seven supporting charts (bar, line, donut, and pie visualizations) was tested by comparing a handful of individual data points against manually filtered values in Excel to confirm the aggregations were correct.

Dashboard Actions Testing

The Keep Only / Exclude action was tested on multiple states, not just Kerala, to confirm that every chart on the dashboard updated consistently regardless of which state was selected.

Story Validation

Each story point was clicked through in sequence to confirm the correct dashboard view and caption appeared for that point, and that navigating backward and forward did not break any chart's interactivity.

Publishing Validation

After publishing, the team re-ran every test above directly on the Tableau Public page in a browser to confirm nothing broke during the publish step, since some local features occasionally behave differently once hosted online.

7. Challenges Faced

Data Cleaning

Challenge: Several district-level rows had missing or zero Production values that would have understated state totals.

Solution: These rows were identified using Tableau's null-value indicators and Excel filters, then removed after confirming they were genuine gaps rather than valid zero-production records.

Map Configuration

Challenge: A few state names in the raw dataset did not initially match Tableau's built-in geographic role for India, leaving those states blank on the map. Solution: State names were manually corrected to match Tableau's expected naming (for example fixing inconsistent capitalization and spacing) until every state plotted correctly.

Dashboard Alignment

Challenge: Charts of different sizes initially left uneven gaps between containers, making the dashboard look unbalanced.

Solution: Fixed-width horizontal and vertical containers were used instead of floating objects, which forced every chart to resize proportionally and kept the layout aligned.

Color Selection

Challenge: The first color palette used bright, mismatched colors across charts that made the dashboard feel cluttered.

Solution: The palette was narrowed down to a consistent cream-and-green theme across all charts, which matched the agricultural subject matter and made the dashboard easier to read.

Dashboard Actions

Challenge: The first version of the filter action affected the map itself, which caused other states to disappear entirely instead of just being de-emphasized. Solution: The action was reconfigured to exclude the source sheet (the map) from the target list, so the map keeps showing every state while only the supporting charts filter down.

Publishing

Challenge: An early publish attempt failed because the workbook referenced a local Excel file that Tableau Public could not access online. Solution: The data source was set to an extract (a saved copy of the data embedded in the workbook) before publishing, which resolved the dependency and let the dashboard load correctly for any online viewer.

8. Project Outcomes

Skills Learned

- Connecting and cleaning real-world Excel data for use in Tableau.
- Building calculated fields (SUM, AVG) to power KPI cards and measures.
- Configuring a geographic map visualization with color-encoded production data.
- Setting up dashboard actions for cross-filtering between charts.
- Designing a multi-chart dashboard layout using containers for consistent alignment.
- Building a Tableau Story to present findings in a guided, business-friendly narrative.
- Publishing and verifying a workbook on Tableau Public.

Knowledge Gained

Beyond the technical Tableau skills, the team gained a clearer understanding of how Indian agricultural production and yield have evolved over more than two decades, which states dominate national output, and how seasonal patterns shape total production.

Dashboard Quality

The final dashboard combines a clean, uncluttered layout with genuinely useful interactivity, meeting the goal of presenting nine distinct visualizations on a single screen without overwhelming the viewer.

Interactive Analytics and Real-time Decision Support

Because every chart is linked through the dashboard action, a user exploring the dashboard - whether a student, a policymaker, or an agri-business analyst - can move from a national overview to a specific state's detailed picture in seconds, supporting faster, more informed decisions than a static report would allow.

Benefits

The project demonstrates, in a compact and practical form, how a well-designed Tableau dashboard can turn 24 years of raw agricultural records into an accessible tool for spotting trends, comparing states, and communicating insights to a non-technical audience.

9. Development Summary

The project moved through a clear, structured development process - starting with dataset collection and cleaning, moving through individual chart development, dashboard assembly, interactivity via dashboard actions, story creation, and finally publishing to Tableau Public. Each stage was tested before moving to the next, which kept errors from compounding as the dashboard grew more complex.

Project Success

The finished dashboard successfully brings together nine visualizations - two KPI cards, an interactive India map, and six supporting charts - into one interconnected view, published and verified as fully functional on Tableau Public.

Objectives Achieved

- An interactive, single-screen dashboard summarizing 1997-2021 agricultural data was delivered.
- State, year, season, and crop-level exploration is possible without any coding by the viewer.
- Key patterns - leading states, seasonal contribution, and yield trends - are visible at a glance.
- Real Tableau development skills were practiced end-to-end within a genuine internship deliverable.
- The dashboard was published and confirmed accessible to anyone with the shared link.

Learning Experience

Working as a three-member team on a shared Tableau workbook taught the group how to divide chart-building work while keeping a single, consistent dashboard design, and how much iteration is genuinely needed to get colors, layout, and interactivity feeling polished rather than just functional.

Future Improvements

- Adding a crop-level drill-down so a user could filter by individual crop, not just state and season.
- Including year-over-year growth-rate calculations alongside the raw production figures.
- Extending the dataset beyond 2021 as newer agricultural records become available.

Conclusion

Overall, the India's Agricultural Crop Production Analysis dashboard met every objective set out at the start of the project: it is interactive, visually clear, published, and genuinely useful for exploring more than two decades of Indian agricultural data. The experience strengthened the team's practical Tableau skills and gave a solid foundation for tackling larger analytics projects in the future.